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Miocene clockwise rotation of southwest Japan and formation of curvature of the Median Tectonic Line: Paleomagnetic implications

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Abstract. The origin of the curvature of the Median Tectonic Line (MTL) in southwest Japan is still controversial. In order to investigate its formation, we carried out a paleomagnetic study in the Nohi area, north of the curved part of the MTL in the eastern part of southwest Japan. More than 420 paleomagnetic samples were collected at 35 sites from Cretaceous to Paleogene welded tuffs in the Kasagatake Rhyolites, Oamamiyama Group, and Nohi Rhyolite. Characteristic directions with high unblocking temperature component above 560°C were isolated from 32 sites. Paleomagnetic directions of the Kasagatake Rhyolites and Oamamiyama Group (55–66 Ma) have a clockwise deflected declination (D = 51.7°, I = 52.6°, α_{95} = 8.6°) which is consistent with the characteristic Paleogene paleomagnetic direction of the central part of southwest Japan. A similarly deflected declination is also observed in the peripheral part of the Nohi Rhyolite region, far away from the Atera fault system. Paleomagnetic data indicate that the Nohi area experienced a clockwise rotation through more than 45° with respect to the Asian continent as a part of southwest Japan but that the amount of rotation of the Nohi area is 23° smaller than that of the central part of southwest Japan. A new model is developed to explain the tectonic rotation. At about 15 Ma during the latest stage of the clockwise rotation of southwest Japan, the eastern end moved southward by more than 60 km along a left-lateral strike-slip fault which comprised the present north-south segment of the MTL and the Akaishi Tectonic Line. We conclude that the curvature of the MTL is a composite of the 15 Ma strike-slip fault system and a small bend of the MTL associated with the small differential rotation of the Nohi area with respect to the central part of southwest Japan.

1. Introduction

Clockwise deflected paleomagnetic directions in Late Cretaceous to Paleogene rocks of southwest Japan (SW Japan) were first reported by *Kawai et al.* [1961]. Later paleomagnetic studies established that SW Japan experienced clockwise rotation in Miocene times [e.g., *Otofuji*, 1996]: More than 40° of rotation is claimed and the climax of rotation is dated to 15 Ma. Biostratigraphic and chronological studies of Ocean Drilling Program (ODP) cores from the Japan Sea confirm that the middle Miocene rotation of SW Japan is associated with the opening of the Japan Sea [*Burckle et al.*, 1992; *Ingle*, 1992; *Kaneoka et al.*, 1992].

Southwest Japan was treated as a single tectonic block for a first-order reconstruction of the Japan Sea–Japan arc system [*Otofuji and Matsuda*, 1983; *Jolivet et al.*, 1994]: SW Japan was assumed to be rotated about a pole at 129°E, 34°N. Recent paleomagnetic studies at both ends of SW Japan show the differential rotation within the SW Japan block (Figure 1). North Kyushu Island, the western end of SW Japan, was subjected to a smaller clockwise rotation than the San'in district, the central part of SW Japan [*Ishikawa*, 1997]. Northerly declinations are seen in Cretaceous to Miocene rocks at the Nohi area, the eastern end of SW Japan [*Itoh*, 1988; *Hayashida*, 1994]. *Itoh* [1988] claimed that the eastern part was rotated counterclockwise with respect to the central part of SW Japan after SW Japan had experienced clockwise rotation. More detailed paleomagnetic studies are required to establish a more refined rotation model of SW Japan.

The curvature of the Median Tectonic Line (MTL) is a clue to the higher-order approximation of the tectonics for SW Japan (Figure 1). The MTL is defined as a fault system between the Ryoke granitic body and the Sanbagawa metamorphic body. The MTL is linear in the central part of SW Japan, with a trend N80°E, whereas the trend of the MTL swings to the north at the eastern part of southwest Japan. The curvature of the MTL appears to reflect bending of an originally more linear fault belt associated with differential rotation within SW Japan during the large Miocene clockwise rotation. Paleomagnetism can provide information on rotation around the vertical axis during the bending of the MTL in SW Japan.

We focused on a paleomagnetic study of the Nohi area north of the curved part of the MTL in the eastern part of SW Japan

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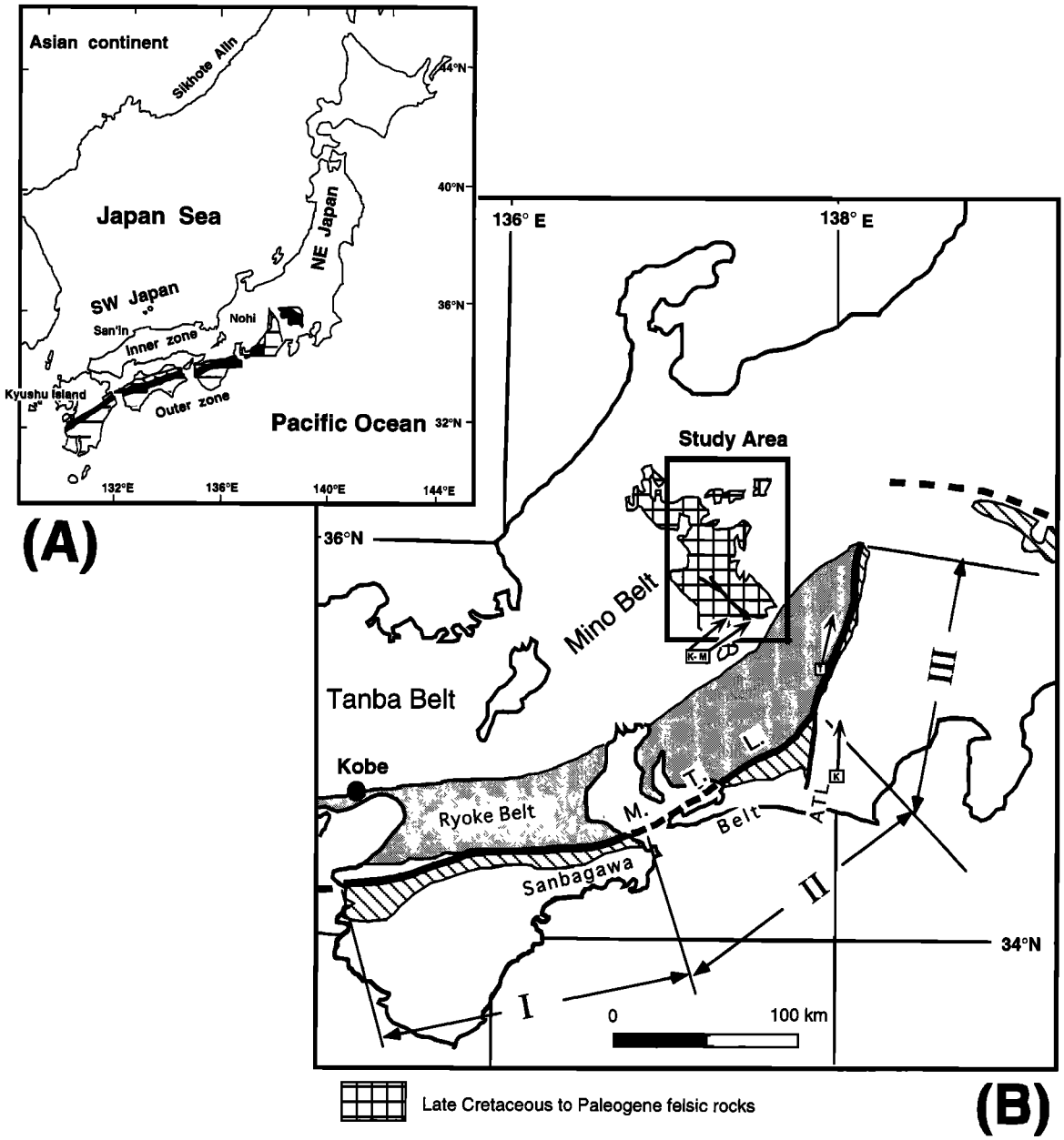


Figure 1. (a) Sketch map of configuration of the Japan arc and the zonally arranged outer zone of SW Japan (the Sanbagawa Belt, Chichibu Belt, and Shimanto Belt from north to south). A significant change is observed in trend of the lineation of the zonal structure of the outer zone in the central part of the Japan arc. (b) Map of study area of the eastern part of SW Japan. The Median Tectonic Line (MTL, thick line) is the boundary between the Ryoke Belt and the Sanbagawa Belt. The MTL is composed of three roughly linear segments of different trend in the eastern part of SW Japan, here recognized as the western segment (N075°–085°E) (segment I), the central segment (N55°E) (segment II), and the eastern segment (N10°E) (segment III). Small arrows are observed paleomagnetic directions of early Middle Miocene for the Kani-Mizunami (K-M), Tomikusa (T), and Kakegawa (K) areas [*Hayashida*, 1986, 1992, 1994; *Hayashida et al.*, 1991]. ATL denotes the Akaishi Tectonic Line.

to understand the tectonics which formed the large change in trend of the MTL (Figure 1). Late Cretaceous to Paleocene welded tuffs of the Kasagatake Rhyolites, the Oamamiyama Group, and the Nohi Rhyolites were the targets of this study (Figure 2).

2. Geological Setting and Sampling

The southwestern part of the Japan arc is divided into the inner and outer zones by the MTL (Figure 1). The inner zone comprises the Hida Belt, Hida Marginal Belt, Mino-Tanba Belt, and Ryoke Belt, while the outer zone is divided into the Sanbagawa, Chichibu, and Shimanto Belts [*Geological Survey of Japan*, 1982]. The Hida belt consists of pre-Jurassic metamorphic and granitic rocks, which are interpreted as a fragment of continental basement [*Hiroi*, 1981]. The other belts are accretionary complexes formed during Late Paleozoic to Paleogene time. Since the Cretaceous, strike-slip movement has occurred along the MTL, which marks the present boundary

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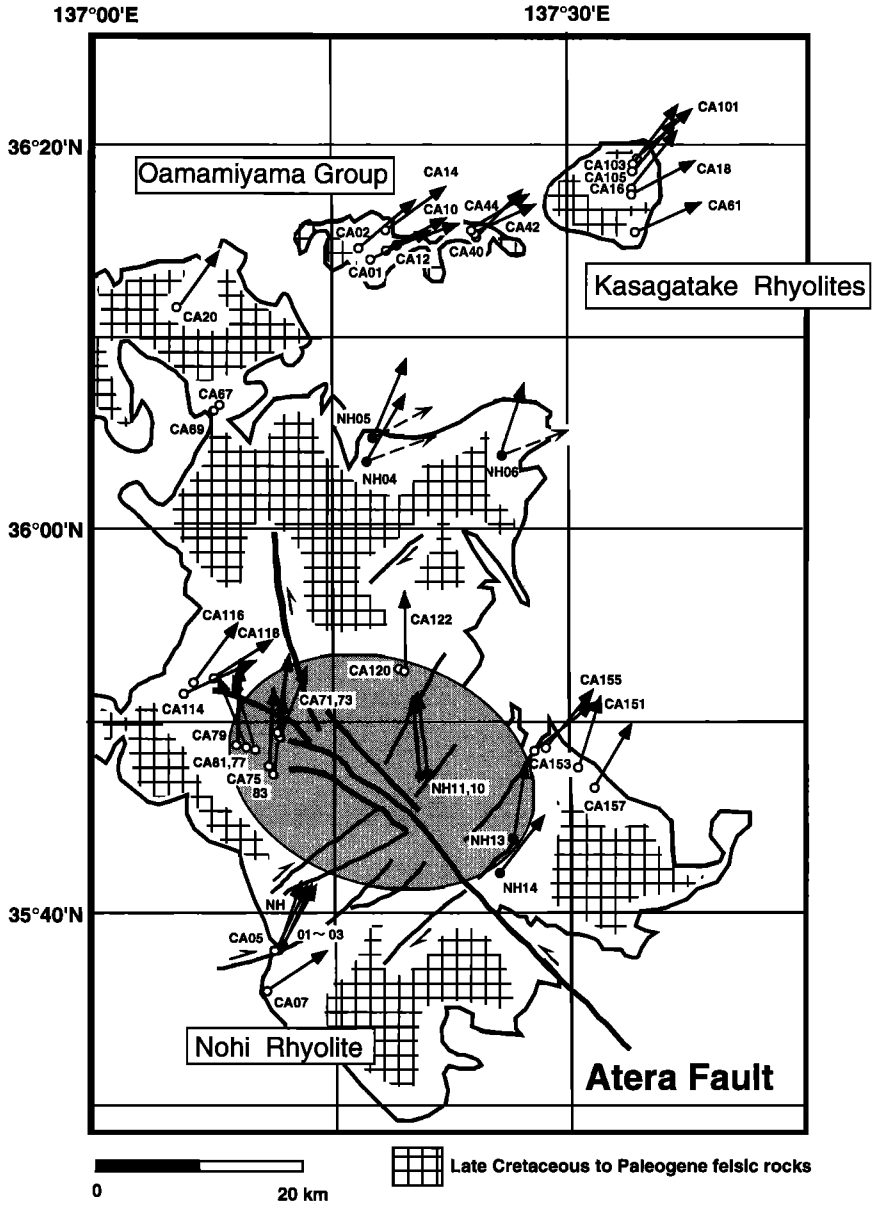


Figure 2. Map of study area of the Nohi area showing location of sampling sites for paleomagnetism. Arrows indicate the paleomagnetic directions for the sampling sites. The paleomagnetic directions of the Nohi Rhyolite are categorized into two groups: Slightly deflected declinations (less than 15° from the north) are confined in a shaded area, whereas large and clockwise deflected declinations are observed outside the shaded area. Open circles (CA) are from the present study; solid circles (NH) are from data of *Itoh* [1988]. The dashed arrows are the paleomagnetic direction before tilt correction previously found by *Itoh* [1988].

between the Ryoke and Sanbagawa Belts [*Ichikawa*, 1980]. Between 132°E and 138°E in longitude, the MTL is composed of three roughly linear segments of different trend (Figure 1), here described as segment I (N75° to 85°E), segment II (N55°E), and segment III (N10°E).

Intense felsic igneous activity occurred during Late Cretaceous to Paleogene time in the Nohi area of the inner zone [*Harayama et al.*, 1985; *Koido*, 1991]. The Nohi Rhyolite accumulated mainly in a large volcanic depression 120 km long by 40 km wide, extending in a NNW-SSE direction from the southern part of the Hida Belt to the northern margin of the Ryoke Belt. To the northeast of the depression, latest Cretaceous to Paleocene volcanic activity yielded the Kasagatake Rhyolites and Oamamiyama Group. These rhyolites and group are composed chiefly of many sheets of welded pyroclastics.

The Late Cretaceous to Paleogene igneous activity is divided into five stages, I to V in ascending stratigraphic order [*Harayama et al.*, 1985]. The Nohi Rhyolite was formed during stages I to III, whereas the Kasagatake Rhyolites and Oamamiyama Group are restricted to stage III. The volcanic rocks of stage II are intruded by the younger granitic rocks of the Ryoke Belt (80 to 65 Ma [*Nakajima et al.*, 1990]). Stage III rocks overlie these granites but are intruded by Paleogene granitic rocks related to stage III.

Ages between 55 to 63 Ma were obtained for the Kasagatake Rhyolites by using whole rock Rb-Sr isochron, K-feldspar

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K-Ar, and zircon fission track methods [*Harayama*, 1990]. The Oamamiyama Group was dated to 66.3 ± 2.0 Ma by the sanidine K-Ar method [*Shibata and Uchiumi*, 1992], which is consistent with an age assignment to the Maastrichtian using fossil flora (e.g., *Aquilapollenites kasaharane*) [*Takahashi and Shimono*, 1982]. Concordant ages determined by different methods confirm the reliability of ages for the Kasagatake Rhyolites and the Oamamiyama Group. Radiometric ages of the Nohi Rhyolite of stages I to III assign it to the period between 65 Ma and 83 Ma using the biotite and sanidine K-Ar, zircon fission track, and whole rock Rb-Sr isochron methods [*Okamoto et al.*, 1975; *Shirahase*, 1982; *Yamada and Kobayashi*, 1988; *Harayama*, 1990; *Koido*, 1991; *Shibata and Uchiumi*, 1992; *Wakita and Koido*, 1994]. Since several rocks of stage II in the Nohi Rhyolite yielded ages younger than the younger granite rocks of the Ryoke Belt [*Shibata and Uchiumi*, 1992; *Yamada et al.*, 1992], resetting of the age may have occurred in some stage II rocks during thermal events occurring after deposition of pyroclastics.

Volcanism was reestablished during latest Pliocene time. Quaternary lavas and pyroclastics overlie the eastern part of the Nohi and Kasagatake Rhyolites.

A strike-slip fault system is developed in the Nohi Rhyolite area. The main fault is the Atera fault, which is a northwest trending left-lateral strike-slip fault with a length of 70 km. The rate of the Holocene left-lateral strike-slip is estimated to be 2–7 mm per year from the offset of stream channels [*Sugimura and Matsuda*, 1965; *Research Group for Active Faults of Japan*, 1991]. Northwest striking faults in this system are left-lateral faults, whereas conjugate strike-slip faults with a northeast trend are right-lateral faults.

We focused the paleomagnetic study on welded tuffs in the Kasagatake Rhyolites (55–63 Ma), the Oamamiyama Group (66 Ma), and the Nohi Rhyolite of stages II and III (65–83 Ma) which are distributed from 35°36′N to 36°18′N in latitude and from 137°05′E to 137°34′E in longitude (Figure 2). More than 420 samples were collected from 35 sites. These areas have been geologically mapped in detail [*Collaborative Research Group for the Nohi Rhyolite*, 1973, 1976, 1979; *Kasahara*, 1979; *Yamada et al.*, 1985, 1989; *Yamada and Wakita*, 1990; *Harayama*, 1990; *Wakita and Koido*, 1994].

The welded tuffs were collected as hand samples oriented by magnetic compass for paleomagnetism. Each paleomagnetic site typically comprised 12 independently oriented samples, distributed over distances ranging up to 15 m. The foliated structure (foliation of stretched pumice and aligned phenocrysts) was identified at each outcrop of the welded tuffs, and this was used to make a correction for postemplacement tilting. Uncertainty arises in measurements of dip (<5°) and strike (~5°) for estimating of the paleohorizontal plane from the foliated structure in the welded tuffs.

3. Paleomagnetism

In the laboratory, individual specimens, 25 mm in diameter and 22 mm long, were prepared from these samples. Natural remanent magnetizations (NRMs) were measured with either a spinner magnetometer (Natsuhara SMM-85) or a cryogenic magnetometer (ScT and 2G) according to the intensity of the magnetization. Specimens were thermally demagnetized using a laboratory-built furnace: The residual magnetic field in which the specimens were cooled is less than 10 nT. Alternating field (AF) demagnetization was carried out with a three-axis tumbler system in three cylindrical mu-metal shields. Pilot specimens show that the thermal demagnetization method was more effective than the AF demagnetization for erasing the secondary magnetization (Figure 3), so that all the remaining specimens were subjected to stepwise thermal demagnetization up to 650°C. Results for each specimen were plotted on orthogonal vector diagrams [*Zijderveld*, 1967] to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis [*Kirschvink*, 1980] was used to estimate the component directions. Paleomagnetic results are listed in Table 1.

3.1. Demagnetization

3.1.1. Kasagatake Rhyolites. NRM intensity ranges between 0.33 and 31 mA m^{-1}. Almost all specimens except for site CA61 reveal three-component magnetization (Figures 3a and 3b). Very clear three-component magnetization is observed in specimens of two sites (CA16 and CA105), which have a NRM intensity of 0.33–4.9 mA m^{-1}. The highest-temperature component reveals straightforward demagnetization behavior with an unblocking temperature between 560°C and 590°C, indicating that this component resides in titanium-poor magnetite. The middle-temperature component appears over the temperature range of 150°C–500°C after a low-temperature component has been erased by 300°C. The direction of the middle-temperature component is roughly antipodal to that of the highest-temperature component. Specimens from site CA61, which have an NRM intensity of 1.0–17 mA m^{-1}, show single-component magnetization with an unblocking temperature between 560°C and 590°C.

3.1.2. Oamamiyama Group. Demagnetization behavior depends strongly on NRM intensity of specimens. Specimens of site CA42 with the strongest NRM intensity (26–850 mA m^{-1}) reveal single-component magnetization (Figure 3c), while specimens with a moderate NRM intensity of 11–170 mA m^{-1} show two-component magnetization. After a low-temperature component is erased by 300°C, a high-temperature component appears over the temperature range of 300°C–590°C. The weakest specimens (3.1–11 mA m^{-1}) of two sites (CA40 and CA44) reveal three-component magnetization (Figure 3d).

3.1.3. Nohi Rhyolite. Two-component magnetization is observed for specimens of 13 sites. The low-temperature component is, in general, erased by 350°C (Figure 3e). Some specimens from site CA114 preserve the low-temperature component up to 500°C. The highest-temperature component reveals straightforward demagnetization behavior with an unblocking temperature between 590°C and 620°C. Both magnetite and hematite probably carry the stable remanent magnetization with almost the same direction.

Three-component magnetization is observed from specimens collected from the peripheral part of this rhyolite region (CA05, CA151, CA153, CA155, CA157) (Figures 3f and 3g). The highest-temperature component appears above 500°C. The unblocking temperature of this component is between 560°C and 590°C, indicating that it resides in magnetite.

Specimens of two sites (CA67 and CA69) have weak NRM intensity which ranges between 0.15 and 3.0 mA m^{-1}. Specimens from these sites reveal variable demagnetization behavior within one site; specimens with weak NRM intensity less than 1.0 mA m^{-1} revealed only spurious remanent magnetization (Figure 3h), whereas single- and two-component magnetization is observed for specimens with higher intensities.

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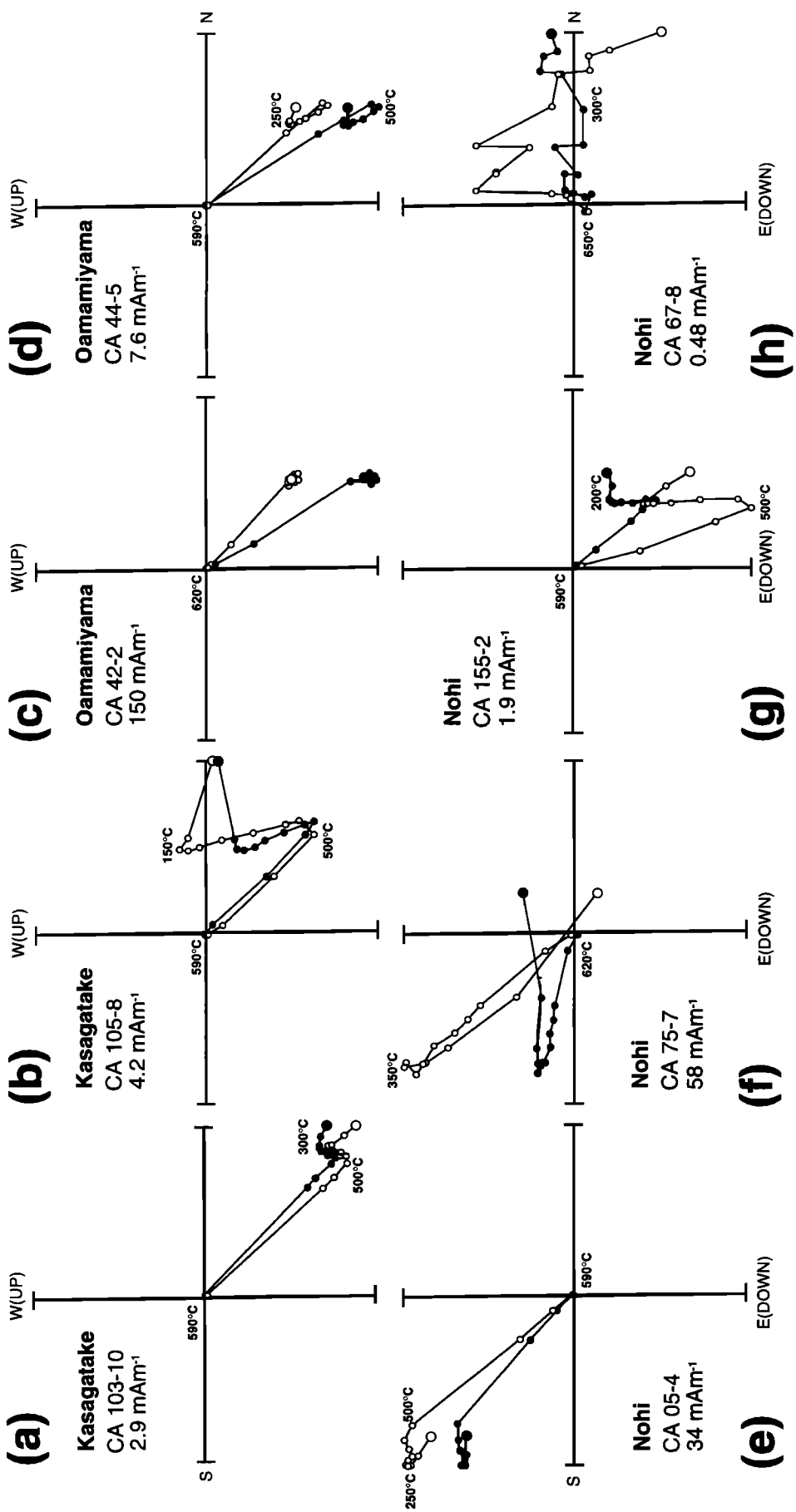


Figure 3. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens of (a, b) the Kasagatake Rhyolites (CA103, CA105), (c, d) the Oamamiyama Group (CA42, CA44), and (e–h) the Nohi Rhyolite (CA05, CA75, CA155, CA67). Solid symbols are for horizontal projection; open symbols are for vertical projection. Tectonic tilt correction has not been applied.

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Table 1. Paleomagnetic Results for the Late Cretaceous to Paleogene Welded Tuffs in the Nohi Area

Site	N	In Situ		Tilt Corrected				Locality	
		D, deg	I, deg	D, deg	I, deg	α_{95}, deg	k	Latitude	Longitude
Kasagatake Rhyolites (55–63 Ma)									
CA101 (HT)	12	41.4	41.4	48.4	40.7	2.0	470.8	36°19′09″N	137°34′15″E
CA103 (HT)	10	36.1	35.1	37.4	32.9	1.9	652.6	36°18′54″N	137°34′15″E
CA105 (HT)	12	41.6	40.5	38.3	32.6	10.1	19.6	36°18′35″N	137°34′15″E
CA18 (HT)	12	252.0	−61.2	242.2	−53.3	9.1	23.8	36°17′27″N	137°34′19″E
CA16 (HT)	12	33.7	35.9	35.3	32.7	9.4	22.1	36°17′27″N	137°34′17″E
CA61 (HT)	4	258.1	−55.5	244.3	−49.6	6.8	181.6	36°15′27″N	137°34′45″E
Mean									
In Situ	6	46.9	45.4			14.1	23.6	36°00′00″N	137°00′00″E
Tilt corrected	6			45.8	40.1	11.1	37.0	36°00′00″N	137°00′00″E
Oamamiyama Group (66 Ma)									
CA42 (HT)	12	56.9	28.0	64.3	41.1	2.4	315.9	36°16′02″N	137°24′01″E
CA44 (HT)	12	51.7	35.3	57.9	44.9	2.5	310.4	36°16′02″N	137°24′01″E
CA40 (HT)	12	52.0	51.3	42.5	66.4	3.0	212.1	36°15′44″N	137°24′35″E
CA14 (HT)	12	232.9	−70.0	233.5	−67.0	3.3	176.9	36°15′25″N	137°18′36″E
CA12 (HT)	12	249.0	−59.5	251.0	−67.4	3.6	144.3	36°14′28″N	137°19′03″E
CA02 (HT)	12	246.3	−73.7	229.2	−75.4	3.0	204.9	36°14′28″N	137°16′56″E
CA10 (HT)	10	247.9	−60.6	243.4	−65.0	3.7	175.8	36°14′23″N	137°18′41″E
CA01 (HT)	9	244.3	−66.1	244.3	−66.1	4.5	129.1	36°13′45″N	137°17′20″E
Mean									
In situ	8	58.6	55.5			11.9	22.8	36°00′00″N	137°00′00″E
Tilt corrected	8			58.9	61.7	8.8	40.4	36°00′00″N	137°00′00″E
Mean (Kasagatake Rhyolites and Oamamiyama Group) (36.0°N, 137.0°E)									
In situ	14	52.9	51.3			8.7	21.7		
Tilt corrected	14			51.7	52.6	8.6	22.5	47.8°N*	211.3°E*
Nohi Rhyolite (65–83 Ma), Peripheral Area									
CA20 (HT)	12	296.6	−80.4	215.0	−73.4	3.0	204.4	36°11′48″N	137°05′17″E
CA118 (HT)	10	255.7	−46.9	237.2	−43.6	10.4	22.6	35°51′59″N	137°07′39″E
CA116 (HT)	10	36.4	54.3	37.4	47.3	9.6	26.3	35°51′45″N	137°06′19″E
CA114 (HT)	8	240.6	−29.5	245.1	−29.8	7.1	61.7	35°51′29″N	137°06′01″E
CA155 (HT)	12	37.6	63.6	38.7	40.6	2.9	224.7	35°48′29″N	137°28′23″E
CA153 (HT)	12	57.5	67.9	49.6	43.4	2.0	452.6	35°48′15″N	137°27′36″E
CA151 (HT)	11	17.3	44.0	17.4	65.0	3.9	137.5	35°47′54″N	137°30′33″E
CA157 (HT)	12	35.7	64.9	29.6	43.3	4.1	127.0	35°46′37″N	137°31′26″E
CA05 (HT)	12	214.9	−43.8	206.3	−54.6	2.6	288.0	35°38′24″N	137°11′26″E
CA07 (HT)	11	230.3	−58.2	235.9	−65.7	2.7	294.6	35°36′02″N	137°11′03″E
Mean									
In situ	10	47.3	57.2			11.6	18.4	36°00′00″N	137°00′00″E
Tilt corrected	10			43.0	51.5	10.1	24.0	36°00′00″N	137°00′00″E
Nohi Rhyolite (65–83 Ma), Area Close to the Atera Fault									
CA122 (HT)	12	163.6	−34.2	179.1	−36.6	1.8	608.0	35°52′34″N	137°19′39″E
CA71 (HT)	12	190.0	−56.1	187.9	−49.9	2.5	303.5	35°50′16″N	137°11′32″E
CA73 (HT)	12	190.3	−52.7	201.3	−55.1	4.1	114.8	35°49′42″N	137°11′26″E
CA81 (HT)	12	163.9	−62.2	158.6	−49.7	8.8	25.3	35°49′30″N	137°09′17″E
CA79 (HT)	12	215.8	−54.9	181.4	−56.0	2.6	287.6	35°49′28″N	137°08′43″E
CA77 (HT)	11	175.7	−54.0	165.6	−74.5	5.7	64.3	35°49′07″N	137°09′53″E
CA75 (HT)	12	190.6	−44.7	184.9	−52.1	3.4	201.9	35°48′29″N	137°11′03″E
CA83 (HT)	12	189.6	−46.2	187.1	−56.5	3.3	171.4	35°47′34′N	137°11′12″E
Mean									
In situ	8	4.5	51.9			9.4	35.8	36°00′00″N	137°00′00″E
Tilt corrected	8			1.4	54.6	8.9	39.6	36°00′00″N	137°00′00″E
Nohi Rhyolite (65–83 Ma), No Reliable Remanent Magnetization									
CA67								36°06′45″N	137°07′49″E
CA69								36°06′38″N	137°07′15″E
CA120								35°52′43″N	137°19′26″E

The magnetization directions of the highest-temperature component (HT) are listed. N is the number of field oriented samples; D and I are declination and inclination, respectively; k is the Fisher precision parameter; α_{95} is the radius of cone of 95% confidence.

*Values shown are for VGP; $A_{95} = 9.0°$, $k = 20.3$.

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Two characteristic directions appear in the high-temperature component in these sites. We recognize that these sites preserve no definitive characteristic direction. It should be noted that these sites are located within 100 m of the Kiyomi granite body, so that original remanent magnetization may have been destroyed by hydrothermal events during intrusion of the granite.

3.2. Paleomagnetic Direction

Paleomagnetic directions of the low-temperature components of magnetization removed by 300°C in specimens with the multiple-component magnetization are dominated by a component in the direction of the present Earth's magnetic field before tilt correction. This component is therefore ascribable to viscous remanent magnetization acquired in recent times.

The high-temperature component directions of 32 of 35 sites are clustered tightly for each site (Figure 4 and Table 1): The site mean precision parameters range from 19.6 to 652.6. Three sites in the Nohi Rhyolite revealed no characteristic direction (CA67, CA69, and CA120) and are thus excluded from further paleomagnetic discussion.

3.2.1. Kasagatake Rhyolites and Oamamiyama group. Both normal and reversed polarities are observed in these rock units (Figure 4a). Declination is characterized by clockwise deflection of between 35° and 71° from a north-south direction. Inclination values range from 33° to 67°. Mean paleomagnetic directions for the Kasagatake Rhyolites, the Oamamiyama Group, and a combined data set of both rock units are listed in Table 1.

The mean paleomagnetic directions of the Oamamiyama Group pass the fold test [*McFadden*, 1990] at 95% confidence limits. Positive reversal test is also attained at 95% confidence limits (class C) [*McFadden and McElhinny*, 1990]. The Oamamiyama Group probably preserves primary magnetization.

The mean paleomagnetic direction of the Kasagatake Rhyolites reveals increases in the precision parameter after tilt correction, although the fold test is statistically inconclusive [*McFadden*, 1990; *Watson and Enkin*, 1993]. Since the normal and reversed polarities are present, the paleomagnetic directions are not due to secondary magnetization. The mean paleomagnetic direction based on the combined data set after tilt correction is recognized to be characteristic for these rock units.

3.2.2. Nohi Rhyolite. Both normal and reversed polarities are observed (Figure 4b) in paleomagnetic directions. Previous paleomagnetic data of *Itoh* [1988] are included. Normal polarity declination deflects clockwise by 50° from the north, while reversed polarity declinations are distributed in a wide range between 158° and 245°.

Declinations are plotted on the distribution map of the Late Cretaceous to Paleogene felsic rocks in Figure 2. The paleomagnetic directions are categorized into two groups: Strongly clockwise deflected declination is observed in the peripheral part of the Nohi Rhyolite region far away from the Atera fault system, whereas slightly deflected directions from a south direction (less than 15°) are observed in a restricted area close to the fault system (this area is shaded in Figure 2 and the directions from this area are plotted on the shaded Schmidt net of Figure 4b). The mean paleomagnetic directions for each group reveal an increase in the precision parameter after tilt correction, although the fold test is statistically inconclusive [*McFadden*, 1990; *Watson and Enkin*, 1993].

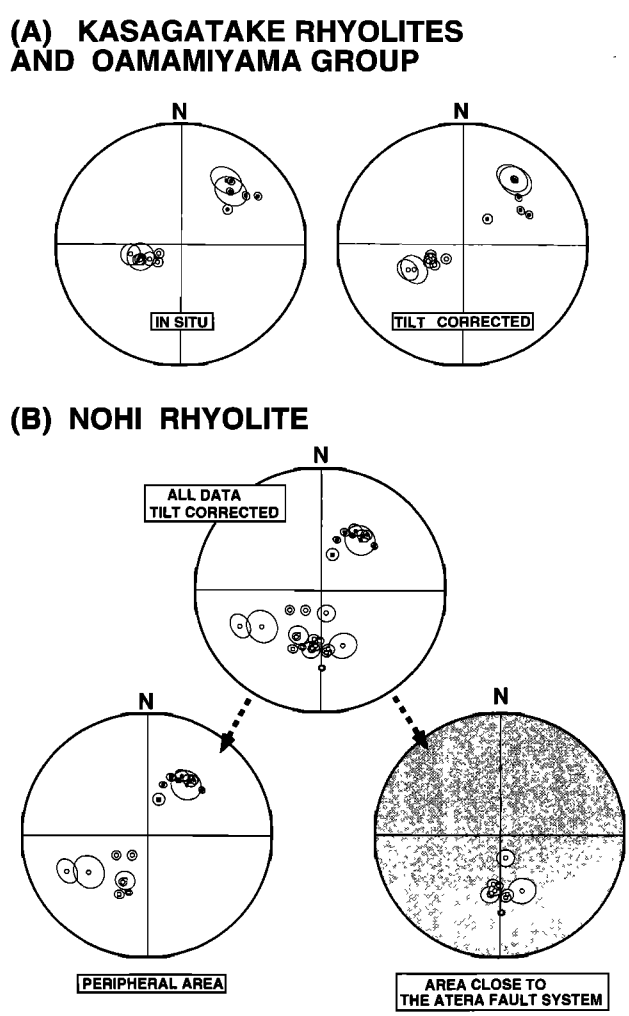


Figure 4. Equal-area projections of the site mean directions (circles) with their 95% cones of confidence. The solid symbols refer to the lower hemisphere, the open symbols refer to the upper hemisphere. (a) The Kasagatake Rhyolites and Oamamiyama Group, before and after tilt correction. Fisher precision parameter k increased from $k = 21.7$ to $k = 22.5$ after tilt correction. (b) The Nohi Rhyolite, after tilt correction. The paleomagnetic directions of the Nohi rhyolite are categorized into two groups: Paleomagnetic directions in the shaded area of Figure 2, which is close to the Atera fault system, reveal slightly deflected declinations from the south, whereas large and clockwise deflected declination is observed outside the shaded area of Figure 2, which is the peripheral part of the Nohi Rhyolite region far away from the fault system. The paleomagnetic directions previously reported by *Itoh* [1988] are shown by squares.

4. Anisotropy of Magnetic Susceptibility

Anisotropy of magnetic susceptibility measurements were performed on all specimens using a Kappabridge KLY-3 apparatus. We calculated the principal directions (maximum, intermediate, and minimum axes: K_1, K_2, and K_3, respectively) for each specimen, and the mean values of the anisotropy degrees ($L = K_1/K_2$ and $F = K_2/K_3$) of three rock units to assess the shape of anisotropy of magnetic susceptibility (Figure 5). Principal axes directions are plotted in stratigraphic coordinates. Because the majority of specimens of the Kasagatake Rhyolites, Oamamiyama Group, and Nohi Rhyolite

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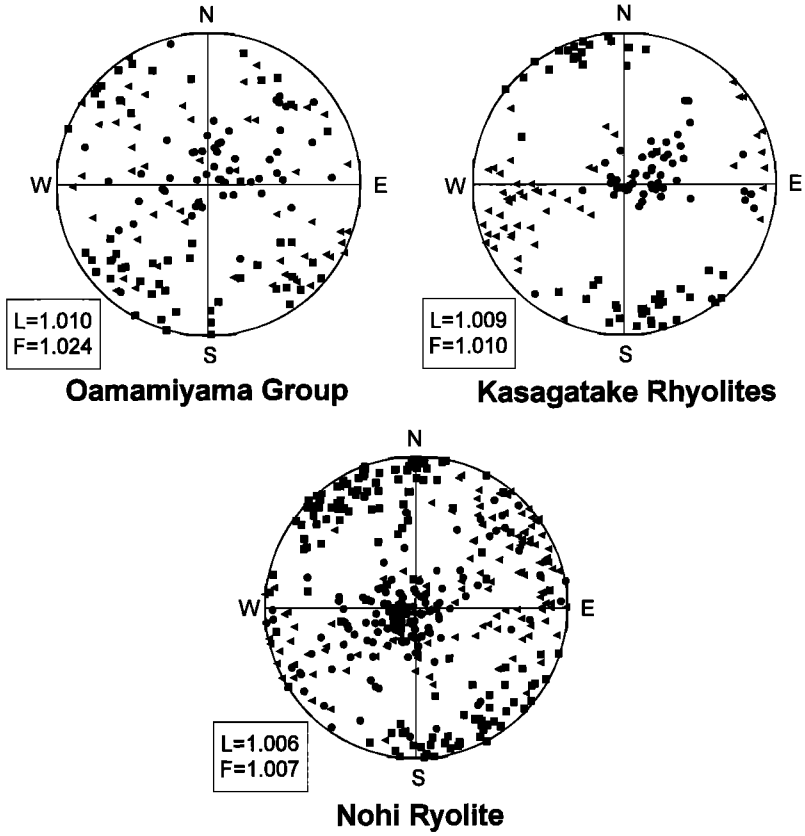


Figure 5. Equal-area projections of the principal directions of susceptibility after tilt correction. Squares are K_1, maximum susceptibility; triangles are K_2, intermediate susceptibility; and circles are K_3, minimum susceptibility. Symbols refer to the lower hemisphere. Mean values of the anisotropy degrees ($L = K_1/K_2$ and $F = K_2/K_3$) are also shown for the Kasagatake Rhyolites, the Oamamiyama Group, and the Nohi Rhyolite.

reveal a weakly foliate nature ($L < F \leqq 1.024$) and their minimum axes are nearly parallel to the vertical axis in stratigraphic coordinate, a depositional origin is suggested for the welded tuffs.

The maximum axes of susceptibility are oblique to declination of the remanent magnetization (Figure 5). The Kasagatake Rhyolites show maximum susceptibility directions of N10°W–S10°E, while the Oamamiyama Group has two directions of N40°W–S40°E and N30°E–S30°W for the maximum axes. These directions of maximum axes are not parallel to the paleomagnetic directions of these rock units (D = 35.3°–71.0°). The maximum axes of the Nohi Rhyolite are distributed between N-S and NW-SE directions, whereas the declination of paleomagnetic directions ranges between N-S and NE-SW. We conclude that directions of remanent magnetization are not controlled by the alignment of anisotropic minerals in welded tuffs.

5. Discussion

5.1. Characteristic Paleomagnetic Direction for Late Cretaceous-Paleocene Time for the Nohi Area

Clockwise deflected directions prevail in the remanent magnetizations of the high temperature component for the Kasagatake Rhyolites and Oamamiyama Group (Figure 2): The combination of these data from two rock units provides a mean direction of D = 51.7°, I = 52.6°, and α_{95} = 8.6° (Figure 4) after tilt correction. The reliability of this direction is ascertained through the positive fold test of the Oamamiyama Group. The presence of normal and reversed polarities indicates that the mean paleomagnetic direction represents a significant time average. Studies of the anisotropy of magnetic susceptibility prove that the paleomagnetic direction has not been affected by pyroclastic flow directions during the formation of the welded tuffs.

A clockwise deflected declination is also observed in the peripheral part of the Nohi Rhyolite region which is outside of the shaded area in Figure 2. The largest clockwise deflection reaches up to 65°. Paleomagnetic directions of both normal and reversed polarities are deflected clockwise from a north-south direction. A mean direction from 10 sites in our study of the peripheral area of the Nohi Rhyolite is D = 43.0°, I = 51.5°, α_{95} = 10.1 (Table 1) after tilt correction, which is compatible with the mean direction of the Kasagatake Rhyolites and Oamamiyama Group. Clockwise deflected declination is probably characteristic of the Nohi Rhyolite. Easterly declinations larger than 60° are also found before tilt correction at sampling sites NH04, NH05, and NH06 in the northern part of the sampling area [*Itoh*, 1988] (Figure 2). These directions should be reexamined by precise measurements of paleohorizontal planes and modern demagnetization techniques.

Slightly deflected directions from the south in the Nohi Rhyolite are confined to the area near the Atera fault system (shaded area in Figure 2). Deflection of the declination from

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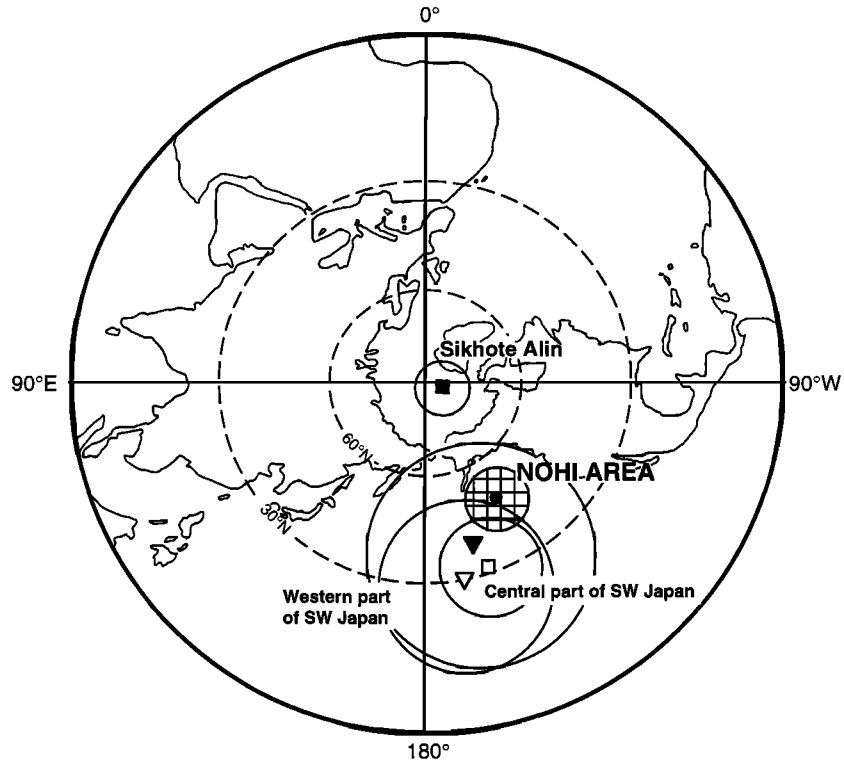


Figure 6. Late Cretaceous to Paleogene poles (with associated cones of 95% confidence) for SW Japan and the Asian continent: The Nohi area (55–66 Ma) (47.8°N, 211.3°E, A_{95} = 9.0°; solid circle, this study); the central part of SW Japan (San'in; 60 Ma) (32.4°N, 199.0°E, A_{95} = 12.4°; open square, [*Otofuji and Matsuda*, 1987]); the western part of SW Japan (north Kyushu Island; 76–97 Ma) (30.6°N, 191.5°E, δm = 20.8°, δp = 19.9°; open triangle [*Ito and Tokieda*, 1986]); (38.8°N, 197.6°E, A_{95} = 28.9°; solid triangle [*Ishikawa*, 1997])), and Sikhote Alin of the Asian continent (50–66 Ma) (84.1°N, 250.5°E, A_{95} = 8.8°; solid square [*Otofuji et al.*, 1995]). This is a polar stereographic projection.

the south ranges between −21.4° and 21.3°. The largest counterclockwise deflection is observed in site CA81 at the northern end of branch faults of the Atera fault. The slightly deflected declination may possibly be attributed to tectonic events (e.g., local rotation and remagnetization) associated with fault movement of the Atera fault system. The northward direction for the Nohi area postulated by *Itoh* [1988] may be due to these slightly deflected declination data.

We conclude that the clockwise deflected declination (D = 51.7°) calculated from the Kasagatake Rhyolites and Oamamiyama Group is an original record of the paleomagnetic direction of Late Cretaceous to Paleocene times (55–66 Ma) for the Nohi area. The characteristic pole position is 47.8°N and 211.3°E (A_{95} = 9.0) (Figure 6). The reliability of the easterly paleomagnetic direction for the Nohi area is confirmed by the large clockwise deflection in declination observed in the early Miocene rocks of the Kani and Mizunami Groups (D = 50° and D = 55°, respectively) south of the southern Nohi area [*Hayashida*, 1986; *Hayashida et al.*, 1991] (Figure 1).

5.2. Clockwise Rotation of the Nohi Area

The clockwise rotation of the paleomagnetic directions of the Nohi area is consistent with those observed in other Cretaceous to Paleocene rocks in SW Japan (Figure 6). The 55–66 Ma pole for the Nohi area falls near the paleomagnetic poles of Cretaceous to Paleogene times for the central part of SW Japan (the welded tuffs of 60 Ma of San'in area (32.4°N, 199.0°E, A_{95} = 12.4°) [*Otofuji and Matsuda*, 1987]) and the western part of SW Japan (the Cretaceous granite (76–97 Ma) of north Kyushu Island (30.6°N, 191.5°E, δm = 20.8°, δp = 19.9°) [*Ito and Tokieda*, 1986]; (38.8°N, 197.6°E, A_{95} = 28.9°) [*Ishikawa*, 1997].

We compare the poles for SW Japan with a Late Cretaceous–Paleogene (50–66 Ma) pole for Sikhote Alin (84.1°N, 250.5°E, A_{95} = 8.8°) [*Otofuji et al.*, 1995], because this pole is recognized as the most reliable Paleogene reference pole for the Asian continent. The pole for the Nohi area, as well as the other poles from SW Japan, is significantly displaced from the pole of Asia. We conclude as a first approximation that the Nohi area was subjected to a large clockwise rotation with respect to the Asian continent as a part of SW Japan. The amount of rotation and its uncertainty ($R \pm \Delta R$) for the Nohi area, and the western and the central parts of SW Japan with respect to Asia, are 45.2° ± 11.7°, 57.5° ± 23.1°, and 67.9° ± 14.2°, respectively. The estimates of the uncertainties quoted above follow *Debiche and Watson* [1995].

The pole position of the Nohi area is distinct from the pole of the central part of SW Japan within 95% confidence limits. When we compare the poles for the Nohi and central part of SW Japan, we claim that the Nohi area experienced clockwise rotation 23.2° ± 14.7° with respect to the central part of SW Japan. This indicates that although the Nohi area was subjected to clockwise rotation, the amount of the rotation was smaller than that of the central part of SW Japan.

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Differential rotation is inferred from curvature of the MTL (Figure 1). More than 65° of bending is observed between segments I and III in the MTL of the eastern part of SW Japan. Segment II, south of the Nohi area, is bent by 20°–30° with respect to segment I, which is an extension of the trend of the central part of SW Japan. This bend is compatible with the amount of rotation expected from the paleomagnetic study (~23°). Tectonic block rotation probably occurred between these two segments.

Since easterly paleomagnetic directions are observed in early Miocene rocks south of the Nohi area [*Hayashida*, 1986; *Hayashida et al.*, 1991], the bending of the eastern part of SW Japan occurred later than the early Miocene, probably during the Miocene clockwise rotation of SW Japan. The curvilinear shape of segment II suggests ductile deformation of the eastern part of SW Japan. Ductile bending has already been inferred from Miocene paleomagnetic data of the Japan Sea side of the eastern part of SW Japan by *Itoh and Ito* [1989].

5.3. Tectonic Implications for the Eastern Part of SW Japan During Its Clockwise Rotation

The paleomagnetically defined vertical axis rotation is unable to explain the whole change in curvature of the eastern part of the MTL, because the paleomagnetism explains only about 35% of the change in curvature. We do not attribute the change in trend between segments II and III of the MTL to tectonic block rotation. The geological nature of segment III of the MTL is an important key to resolving the apparent conflict.

We resolve this disparity by proposing a kinematically permissible model (Figure 7) in which no large differential rotation occurred between the central and eastern parts of southwest Japan during its Miocene clockwise rotation. A postulated left-lateral strike-slip fault zone in SW Japan played an important role in this model during collision with the Izu-Bonin arc in the latest stage of the rotation of SW Japan.

5.3.1. Tectonic description for 20–16 Ma. An originally more linear tectonic fault belt of the proto-MTL divided the SW Japan into the outer and inner zones. A rigid sliver of SW Japan was rotated about a pole around 129°E, 34°N [*Otofuji and Matsuda*, 1983; *Otofuji et al.*, 1991], and the eastern end of SW Japan approached the Izu-Bonin arc. An incipient slip plane formed in the eastern part of SW Japan due to collision with the Izu-Bonin arc.

5.3.2. Tectonic description for 15 Ma. SW Japan was divided into two blocks from a tectonic viewpoint: the Kanto block and the main block. The Kanto block ceased its southward displacement and was independently rotated clockwise through more than 60° with respect to the rest of SW Japan about a pole near the western end of the Kanto block [*Hyodo and Niitsuma*, 1986; *Takahashi and Watanabe*, 1993]. The main block continued to rotate clockwise about a pole around 129°E, 34°N, and the eastern end of the main block moved southward by more than 60 km along a left-lateral strike-slip fault zone. This is now recognized as a linear geomorphologic feature composed of segment III of the present MTL and the Akaishi Tectonic Line (ATL) (Figure 1). The eastern part of this main block behaved as ductile material and bent by about 20° with respect to the central part of SW Japan.

5.3.3. Present tectonic description. Curvature of the present MTL is a composite of the 15 Ma strike-slip fault (segment III) and a small bend of the MTL associated with a small differential rotation of the Nohi area (segment II).

This model requires that segment III of the MTL and the ATL formed a sinistral strike-slip fault which moved during the latest stage of clockwise rotation of SW Japan at about 15 Ma. Continuity of structure from segment III of the MTL to the ATL is clearly manifested as a narrow strip of high Bouguer gravity anomaly [*Gravity Research Group in Southwest Japan*, 1994]. Sinistral shear along segment III of the MTL is indicated by the microstructure of the mylonitic rocks [*Takagi*, 1986; *Yamamoto*, 1994], although it is hard to estimate the amount of displacement. Sinistral displacement of more than 50 km is estimated in the Sanbagawa Belt, Chichibu Belt,

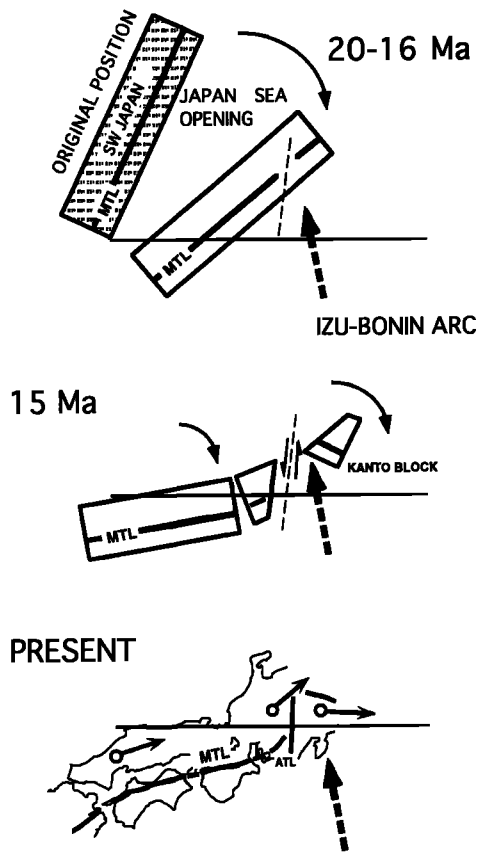


Figure 7. A model of formation of the curved Median Tectonic Line (MTL) associated with the Miocene clockwise rotation of SW Japan. (top) At 20–16 Ma, An originally more linear tectonic fault belt of the proto-MTL divided SW Japan into the outer and inner zones. A rigid sliver of SW Japan was rotated clockwise, and its eastern end approached the Izu-Bonin arc. The dashed line represents an incipient slip plane originating from the collision with the Izu-Bonin arc. (middle) At 15 Ma the Kanto block ceased its southward displacement and was independently rotated clockwise through more than 60° with respect to the remaining part of SW Japan. The remaining block continued to rotate clockwise. The eastern end of this block moved southward by more than 60 km along a left-lateral slip fault zone. The eastern part of this block bent by about 20° with respect to the central part of SW Japan. (bottom) The curvature of the present MTL is an apparent feature which is a composite of the 15 Ma strike-slip fault and a small bend of the MTL associated with small differential rotation of the Nohi area. ATL denotes the Akaishi Tectonic Line. Large arrows are observed Paleocene paleomagnetic directions for the Nohi area of this study and the central part of SW Japan [*Otofuji and Matsuda*, 1987], and the early middle Miocene one of the Kanto block [*Hyodo and Niitsuma*, 1986].

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Sambozan Belt, and Shimanto Belt along the ATL [*Matsushima*, 1973; *Kano et al.*, 1993]. Recent chronological studies confirm the age of 15 Ma as the time of fault movement of these tectonic lines: The fine fraction separated from fault gouges in segment III of the MTL is dated as 18.9 Ma by K-Ar and 17.9 by Rb-Sr methods [*Shibata and Takagi*, 1988], while *Tanaka et al.* [1995] assigned 15 Ma as the age of the micaceous minerals in the fault gouges of the ATL by the K-Ar method. A calcareous foraminifera age of the Futamata group also suggests that the ATL slip motion occurred during the period of zone CN4 (14–16 Ma) [*Kano et al.*, 1993]. Segment III of the MTL and the ATL behaved as a long left-lateral strike-slip fault during the latest stage of the clockwise rotation of SW Japan.

This strike-slip fault seems to have brought about local tectonic rotation in the blocks along the fault. Northerly directions are reported for the late early Miocene to early middle Miocene sedimentary rocks in the Tomikusa area (D = 13°) west of segment III of the MTL and in the Kakegawa area (D = 4°) east of the ATL [*Hayashida*, 1992, 1994] (Figure 1). Both areas are located within 15 km of the 15 Ma sinistral strike-slip fault of segment III of the MTL or the ATL. Since counterclockwise rotation is predicted for these areas from the sense of sinistral motion, the original easterly deflected direction may have been rotated counterclockwise to a northerly direction due to plastic drag by sinistral strike-slip motion.

5.4. Southerly Reversed Polarity Directions Along the Atera Fault

Paleomagnetic directions with reversed polarity from the area close to the Atera fault are tightly clustered and reveal small to zero deflection from the south (D = 181.4°, I = 54.6°, α_{95} = 8.9). One possible explanation for the no-deflected direction from the south is local tectonics which gave rise to counterclockwise block rotation along the Atera fault. Comparison with the paleomagnetic direction of the Kasagatake Rhyolites and Oamamiyama Group requires that counterclockwise block rotation in this area was between 30° and 73°.

Sinistral displacement of the Atera fault system is presumably responsible for this large counterclockwise rotation. The Atera fault splays at its northern end into several branch faults: the Hagiwara, Gero-higashi, Gero-nishi, Miyachi, and Natuyake faults [*Wakita and Koido*, 1994] (Figure 2). Sampling localities CA71, CA73, CA75, CA77, CA79, CA81, and CA83, which experienced the largest counterclockwise rotation, are located among these faults. The remaining localities which experienced counterclockwise rotational motion (CA122, NH10, NH11, NH13 [*Itoh*, 1988]) are located among conjugate faults of the Atera fault. The geological study shows that the Atera fault and its northern branch faults moved sinistrally, whereas the conjugate faults moved dextrally [*Research Group for Active Faults of Japan*, 1991]. Fault blocks in sinistral slip fault zones in the northern part of the fault system should have rotated counterclockwise about vertical axes [*MacDonald*, 1980] (Figure 8), whereas the regions between the two right-lateral faults in the central part should have experienced a counterclockwise rotation due to the left-lateral faulting of the Atera fault [*Kanaori et al.*, 1990]. This fault system probably induced the counterclockwise rotation of sampling localities within this system. Similar large rotation of blocks confined to narrow zone close to a master fault have recently been observed from the Anatolian fault zone [*Piper et al.*, 1997]. Activation of the Atera fault system commenced later than 2 Ma [*Matsuda*, 1976; *Kanaori et al.*, 1992], so the counterclockwise rotation of the Nohi Rhyolite is a reflection of local tectonics under the WNW-ESE compression system of the Quaternary in SW Japan due to E-W maximum horizontal stress from the collision of the Izu Peninsula [*Tsukahara and Ikeda*, 1991; *Le Pichon et al.*, 1996].

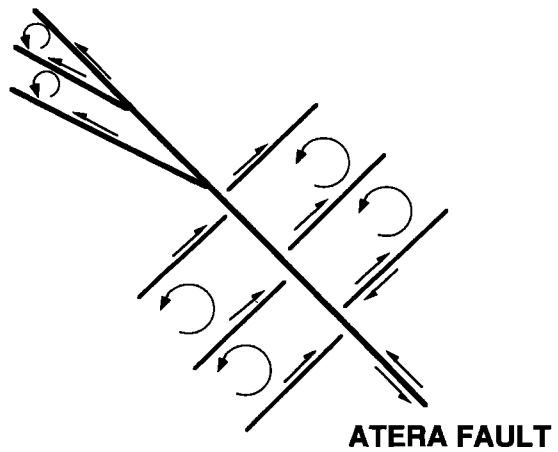


Figure 8. Schematic of the Atera fault system and tectonic rotation model. In the northern part, fault blocks in sinistral slip fault zones rotate counterclockwise about vertical axes [*MacDonald*, 1980]. In the central part, the regions between the two right-lateral faults experience a counterclockwise rotation due to the left-lateral faulting of the Atera fault [*Kanaori et al.*, 1990].

An alternative explanation of the southerly reversed direction is remagnetization. Specimens from the area along the Atera fault show unblocking temperature less than 350°C for the low-temperature component magnetization, whereas the low-temperature component magnetization of many localities from the peripheral area of the Nohi Rhyolite has higher unblocking temperature about 500°C. *Pullaiah et al.* [1975] estimated unblocking temperatures at about 250°C (350°C) for single-domain magnetite (hematite) grains which acquired viscous partial thermoremanent magnetization in the last 1 Ma. It may be possible that the welded tuffs along the fault suffered complete remagnetization during the fault activity of the Atera fault system in the last 2 Ma and later acquired viscous partial thermoremanent magnetization. The remagnetization may have occurred presumably due to fluid migration within a short duration of one reversed polarity period. This mechanism can explain three observations: (1) only reversed polarity, (2) non-deflected direction from the south, and (3) small dispersion in declination. Much work still is needed to better understand the origin of the paleomagnetic direction along this fault.

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References

Burckle, L. H., et al., Biostratigraphic and biochronologic synthesis of Leg 127 and 128: Sea of Japan, *Proc. Ocean Drill. Program Sci. Results*, *127/128*, Part 2, 1219–1228, 1992.

21562202b, 1999, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/1999JB900086 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Collaborative Research Group for the Nohi Rhyolite, Stratigraphy and the volcanic history of the Nohi Rhyolite in its eastern marginal part, *Earth Sci., 27*, 161–179, 1973.

Collaborative Research Group for the Nohi Rhyolite, Stratigraphy and the volcanic history of the Nohi Rhyolite in its western parts, *Earth Sci., 30*, 193–205, 1976.

Collaborative Research Group for the Nohi Rhyolite, Stratigraphy and outline of the volcanic history of the Nohi Rhyolite in its northern part, *Mem. Geol. Soc. Jpn., 17*, 165–176, 1979.

Debiche, M. G., and G. S. Watson, Confidence limits and bias correction for estimating angles between directions with applications to paleomagnetism, *J. Geophys. Res., 100*, 24,405–24,429, 1995.

Geological Survey of Japan, Kokudo map, in *Geological Atlas of Japan*, 119 pp., Tokyo, 1982.

Gravity Research Group in Southwest Japan, A Bouguer gravity anomaly map in central Japan, *Rep. Geol. Surv. Jpn., 280*, 29–36, 1994.

Harayama, S., *Geology of the Kamikochi District With Geological Sheet Map at 1:50,000* (in Japanese with English abstract), 175 pp., Geol. Surv. of Jpn., Tsukuba, 1990.

Harayama, S., Y. Koido, K. Ishizawa, Y. Nakai, and T. Kutsukake, Cretaceous to Paleogene magmatism in the Chubu District, Japan, *Earth Sci., 39*, 344–357, 1985.

Hayashida, A., Timing of rotational motion of southwest Japan inferred from paleomagnetism of the Setouchi Miocene Series, *J. Geomagn. Geoelectr., 38*, 295–310, 1986.

Hayashida, A., Paleomagnetic data from the lower Miocene Tomikusa Group, central Japan, *Bull. Mizunami Fossil Mus., 19*, 67–74, 1992.

Hayashida, A., Paleomagnetism of Miocene sedimentary rocks in the Kakegawa area and implication for tectonic rotation in central Japan, *J. Geomagn. Geoelectr., 46*, 1051–1066, 1994.

Hayashida, A., T. Fukui, and M. Torii, Paleomagnetism of the early Miocene Kani Group in southwest Japan and its implication for the opening of the Japan Sea, *Geophys. Res. Lett., 18*, 1095–1098, 1991.

Hiroi, Y., Subdivision of the Hida metamorphic complex, central Japan and its bearing on the geology of Far East in pre-Sea of Japan time, *Tectonophysics, 76*, 317–333, 1981.

Hyodo, H., and N. Niitsuma, Tectonic rotation of the Kanto Mountains, related with the opening of the Japan Sea and collision of the Tanzawa block since middle Miocene, *J. Geomagn. Geoelectr., 38*, 335–348, 1986.

Ichikawa, K., Geohistory of the Median Tectonic Line of southwest Japan, *Mem. Geol. Soc. Jpn., 18*, 187–210, 1980.

Ingle, J. C., Jr., Subsidence of the Japan Sea: Stratigraphic evidence from ODP sites and onshore sections, *Proc. Ocean Drill. Program Sci. Results, 127/128*, Part 2, 1197–1218, 1992.

Ishikawa, N., Differential rotations of north Kyushu Island related to middle Miocene clockwise rotation of SW Japan, *J. Geophys. Res., 102*, 17,729–17,745, 1997.

Ito, H., and K. Tokieda, Tilting movement of the Japanese islands inferred from Cretaceous and early Tertiary paleomagnetic data, *J. Geomagn. Geoelectr., 38*, 361–386, 1986.

Itoh, Y., Differential rotation of the eastern part of southwest Japan inferred from paleomagnetism of Cretaceous and Neogene rocks, *J. Geophys. Res., 93*, 3401–3411, 1988.

Itoh, Y., and Y. Ito, Confined ductile deformation in the Japan arc inferred from paleomagnetism, *Tectonophysics, 167*, 57–73, 1989.

Jolivet, L., K. Tamaki, and M. Fournier, Japan Sea, opening history and mechanism: A synthesis, *J. Geophys. Res., 99*, 22,237–22,259, 1994.

Kanaori, Y., Y. Endo, K. Yairi, and S. Kawakami, A nested fault system with block rotation caused by left-lateral faulting: The Neodani and Atera faults, central Japan, *Tectonophysics, 177*, 401–418, 1990.

Kanaori, Y., S. Kawakami, and K. Yairi, The block structure and Quaternary strike-slip block rotation of central Japan, *Tectonics, 11*, 47–56, 1992.

Kaneoka, I., Y. Takigami, N. Takaoka, S. Yamashita, and K. Tamaki, ^{40}Ar-^{39}Ar analysis of volcanic rocks recovered from the Japan Sea floor: Constraints on the age of formation of the Japan Sea, *Proc. Ocean Drill. Program Sci. Results, 127/128*, Part 2, 819–836, 1992.

Kano, K., H. Tanaka, T. Yoshida, and N. Matsui, Formation of the Akaishi Tectonic Zone and its significance on the crustal reconstruction of the eastern part of southwest Japan arc during the Miocene, *Mem. Geol. Soc. Jpn., 42*, 203–223, 1993.

Kasahara, Y., Geology of the Oamamiyama Group—Late Cretaceous acid volcanism on the Hida Marginal Belt, central Japan, *Mem. Geol. Soc. Jpn., 17*, 177–186, 1979.

Kawai, N., H. Ito, and S. Kume, Deformation of the Japanese Islands as inferred from rock magnetism, *Geophys. J. R. Astron. Soc., 6*, 124–129, 1961.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699–718, 1980.

Koido, Y., A late Cretaceous-Paleogene cauldron cluster: The Nohi Rhyolite, central Japan, *Bull. Volcanol., 53*, 132–146, 1991.

Le Pichon, X., F. Pollitz, M. Fournier, J. P. Cadet, S. Lallemant, and N. Chamot-Rooke, Distribution of shortening landward and oceanward of the eastern Nankai Trough due to the Izu-Ogasawara ridge collision, *Earth Planet. Sci. Lett., 137*, 145–156, 1996.

MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and structural tilt correction in paleomagnetic studies, *J. Geophys. Res., 85*, 3659–3669, 1980.

Matsuda, T., Empirical rules on sense and rate of crustal movement, *J. Geod. Soc. Jpn., 22*, 252–263, 1976.

Matsushima, N., The median tectonic line in the Akaishi Mountains, in *Median Tectonic Line*, edited by R. Sugiyama, pp. 9–27, Tokai Univ. Press, Tokyo, 1973.

McFadden, P. L., A new fold test for palaeomagnetic studies, *Geophys. J. Int., 103*, 163–169, 1990.

McFadden, P. L., and M. W. McElhinny, Classification of the reversal test in palaeomagnetism, *Geophys. J. Int., 103*, 725–729, 1990.

Nakajima, T., T. Shirahase, and K. Shibata, Along-arc lateral variation of Rb-Sr and K-Ar ages of Cretaceous granitic rocks in southwest Japan, *Contrib. Mineral. Petrol., 104*, 381–389, 1990.

Okamoto, K., S. Nohda, Y. Masuda, and T. Matsumoto, Significance of Cs/Rb ratios in volcanic rocks as exemplified by the Nohi Rhyolite complex, central Japan, *Geochem. J., 9*, 201–210, 1975.

Otofuji, Y., Large tectonic movement of the Japan arc in late Cenozoic times inferred from paleomagnetism; Review and synthesis, *Island Arc, 5*, 229–240, 1996.

Otofuji, Y., and T. Matsuda, Paleomagnetic evidence for the clockwise rotation of southwest Japan, *Earth Planet Sci. Lett., 62*, 349–359, 1983.

Otofuji, Y., and T. Matsuda, Amount of clockwise rotation of southwest Japan—Fan shape opening of the southwestern part of the Japan Sea, *Earth Planet Sci. Lett., 85*, 289–301, 1987.

Otofuji, Y., T. Itaya, and T. Matsuda, Rapid rotation of southwest Japan—Palaeomagnetism and K-Ar ages of Miocene volcanic rocks of southwest Japan, *Geophys. J. Int., 105*, 397–405, 1991.

Otofuji, Y., et al., Late Cretaceous to Early Paleogene paleomagnetic results from Sikhote Alin, far eastern Russia: Implications for deformation of east Asia, *Earth Planet. Sci. Lett., 130*, 95–108, 1995.

Piper, J. D. A., O. Tatar, and H. Gursoy, Deformation behaviour of continental lithosphere deduced from block rotations across the north Anatolian fault zone in Turkey, *Earth Planet. Sci. Lett., 150*, 191–203, 1997.

Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop, Magnetization changes caused by burial and uplift, *Earth Planet. Sci. Lett., 28*, 133–143, 1975.

Research Group for Active Faults of Japan, *Active Faults in Japan: Sheet Maps of Inventories*, revised ed., 437 pp., Tokyo Univ. Press, Tokyo, 1991.

Shibata, K., and H. Takagi, Isotopic ages of rocks and intrafault materials along the Median Tectonic Line—An example in the Bunguitoge area, Nagano Prefecture, *J. Geol. Soc. Jpn., 94*, 35–50, 1988.

Shibata, K., and S. Uchiumi, K-Ar age results, 4, New data from the Geological Survey of Japan, *Bull. Geol. Surv. Jpn., 43*, 359–367, 1992.

Shirahase, T., Rubidium-strontium ages and strontium isotope ratios of the Late Cretaceous felsic volcanic rocks in central Japan, paper presented at 5th International Conference on Geochronology, Cosmochronology, Isotope Geology, Geochem. Soc. of Jpn., Nikko, 1982.

Sugimura, A., and T. Matsuda, Atera fault and its displacement vectors, *Geol. Soc. Am. Bull., 76*, 509–522, 1965.

Takagi, H., Implications of mylonitic microstructure for the geotectonic evolution of the Median Tectonic Line, central Japan, *J. Struct. Geol., 8*, 3–14, 1986.

Takahashi, K., and H. Shimono, Maestrichtian microflora of the Miyadano-gawa Formation in the Hida district central Japan, *Bull. Fac Lib. Arts Nagasaki Univ. Natl. Sci., 22*, 11–188, 1982.

Takahashi, M., and Y. Watanabe, Paleomagnetism of the Miocene

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igneous rocks in the Uchiyama area, central Japan, *J. Geomagn. Geoelectr.*, *45*, 89–101, 1993.

Tanaka, H., N. Uehara, and T. Itaya, Timing of the cataclastic deformation along the Akaishi Tectonic Line, central Japan, *Contrib. Mineral. Petrol.*, *120*, 150–158, 1995.

Tsukahara, H., and R. Ikeda, Crustal stress orientation in the central part of Honshu, Japan—Stress provinces and their origins, *J. Geol. Soc. Jpn.*, *97*, 461–474, 1991.

Wakita, K., and Y. Koido, Geology of the Gero District, with geological sheet map at 1:50,000 (in Japanese with English abstract), 79 pp., Geol. Surv. of Jpn., Tsukuba, 1994.

Watson, G. S., and R. J. Enkin, The fold test in paleomagnetism as a parameter estimation problem, *Geophys. Res. Lett.*, *20*, 2135–2137, 1993.

Yamada, N., and T. Kobayashi, Geology of the Ontakesan District, with geological sheet map at 1:50,000 (in Japanese with English abstract), 136 pp., Geol. Surv. of Jpn., Tsukuba, 1988.

Yamada, N., and K. Wakita, Geological map Iida, scale 1:200,000 Geol. Surv. of Jpn., Tsukuba, 1990.

Yamada, N., M. Adachi, S. Kajita, S. Harayama, H. Yamazaki, and M. Bunno, *Geology of the Takayama District* (in Japanese with English abstract), *Quadrangle Ser.*, scale 1:50,000, 102 pp., Geol. Surv. of Jpn., Tsukuba, 1985.

Yamada, N., T. Nozawa, S. Harayama, F. Takizawa, and H. Katori, Geological Map Takayama, scale 1:200,000, Geol. Surv. of Jpn., Tsukuba, 1989.

Yamada, N., K. Shibata, E. Tsukuda, S. Uchiumi, A. Matsumoto, H. Takagi, and H. Akahane, Radiometric ages of igneous rocks around the Atera fault, central Japan, with special reference to the age of activity of the Atera fault, *Bull. Geol. Soc. Jpn.*, *43*, 759–779, 1992.

Yamamoto, H., Kinematics of mylonitic rocks along the Median Tectonic Line, Akaishi Range, central Japan, *J. Struct. Geol.*, *16*, 61–70, 1994.

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism*, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier, New York, 1967.

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